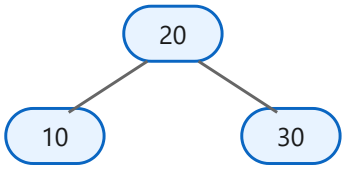
Maps, Sets and Hashing

Lookup by key: `map` and `set` are sorted trees, the unordered containers are hash tables. `operator[]` versus `find`, `try_emplace`, `multimap`, custom comparators and hashes, and `pair` and `tuple`. Assumes STL Containers.

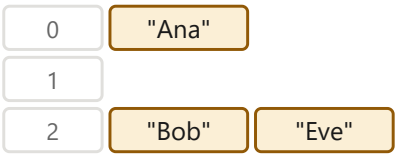
Associative and unordered containers

map / set: balanced tree, sorted, O(log n)



in-order walk = sorted order

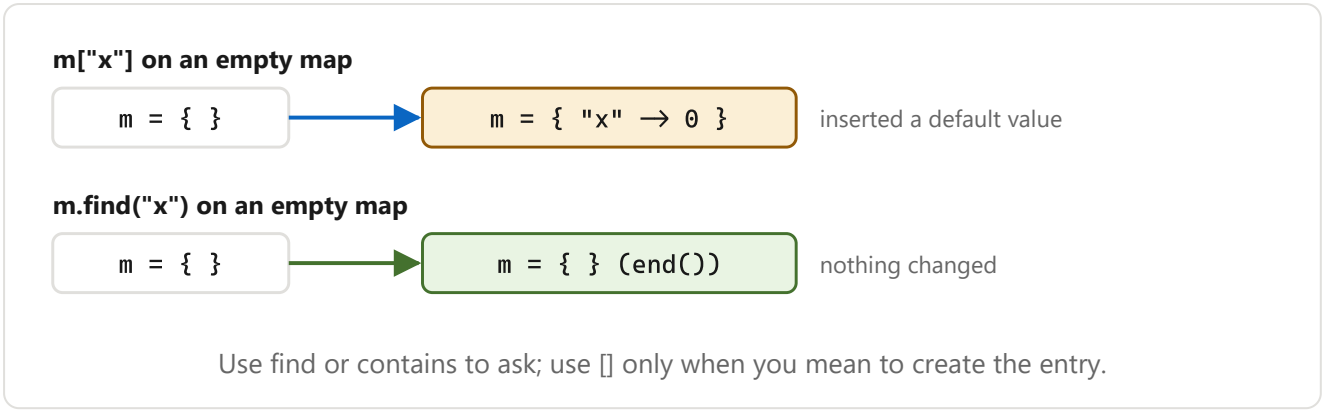
unordered_map: hash table, average O(1)



hash(key) picks a bucket; collisions chain (Bob, Eve)

Need order or range queries: map. Plain lookup: unordered_map.

	`MAP` / `SET`	`UNORDERED_MAP` / `UNORDERED_SET`
Implementation	balanced binary tree (usually red-black)	hash table
Order of iteration	**sorted** by key	unspecified
find / insert / erase	O(log n)	average O(1), worst case O(n)
Key requirement	`operator<` (a strict weak ordering)	a hash function and `operator==`
Range queries (`lower_bound`)	yes	no




```cpp
std::map<std::string, int> age;
// operator[] INSERTS a default (0) if the key is missing
age["Ana"] = 30;
// does nothing if "Bob" already exists
age.insert({"Bob", 25});
age.insert_or_assign("Bob", 26);          // C++17: insert or overwrite

if (auto it = age.find("Ana"); it ≠ age.end())   // find never inserts
    // it→first is the key, it→second the value
    std::cout << it→second;

// structured bindings; sorted by name
for (const auto& [name, years] : age)
    std::cout << name << ' ' << years << '\n';

std::unordered_set<int> seen;
if (!seen.insert(5).second) { /* second == false means it was already there
*/ }
```

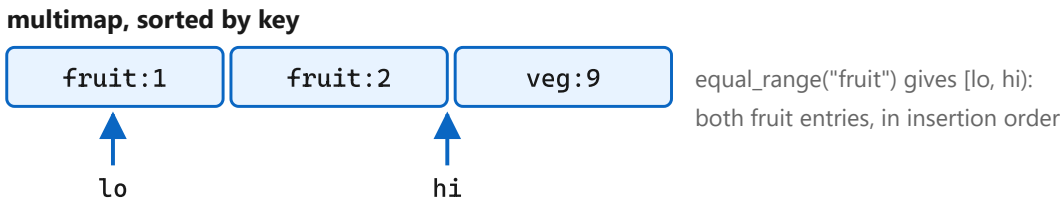
`try_emplace`, `multimap` and `multiset`

CALL	IF THE KEY ALREADY EXISTS
`m[k] = v`	overwrites (and inserts a default first if missing)
`m.insert({k, v})` / `m.emplace(k, v)`	does nothing, but **may already have moved from or built the value**
`m.try_emplace(k, args ... )` (C++17)	does nothing and **guarantees the arguments are not touched**: compiled, a moved-in 40-character string was intact afterwards
`m.insert_or_assign(k, v)` (C++17)	overwrites

```cpp
std::map<int, std::string> a{{1, "old"}};
std::string payload(40, 'p');
// key 1 exists: nothing happens, payload still holds 40 chars
a.try_emplace(1, std::move(payload));
// new key: the string is moved into the map
a.try_emplace(2, std::move(payload));

// duplicate keys allowed
std::multimap<std::string, int> mm{{"fruit", 1}, {"fruit", 2}, {"veg", 9}};
// the range of all entries with that key
auto [lo, hi] = mm.equal_range("fruit");
// 2: there is no operator[] because a key may have many values
mm.count("fruit");
```

- `multimap` and `multiset` allow duplicate keys and keep equal keys in insertion order. There is no `operator[]`.
- `try_emplace` is the right call for "insert if absent" when building the value is expensive or the argument is move-only.



A multimap keeps duplicate keys next to each other.

Custom keys: comparators and hashes

A `map` / `set` needs an **ordering**; an `unordered_map` / `unordered_set` needs a **hash and equality**. For your own key types, you provide them.

```cpp
// an ordering that ignores case
struct NoCase {
    bool operator()(const std::string& a, const std::string& b) const {
        return std::lexicographical_compare(a.begin(), a.end(), b.begin(),
b.end(),
            [](unsigned char x, unsigned char y) { return std::tolower(x) <
std::tolower(y); });
    }
};
```

```cpp
// the comparator is part of the TYPE
std::map<std::string, int, NoCase> m;
// the same key: size is 1 (compiled: 2 keys with "beta")
m["Alpha"] = 1; m["ALPHA"] = 2;

struct Product { std::string shop; int id; bool operator==(const Product&)
const = default; };
struct ProductHash {
    std::size_t operator()(const Product& p) const {
        std::size_t h = std::hash<std::string>{}(p.shop);
        // combine the member hashes
        h ^= std::hash<int>{}(p.id) + 0x9e3779b97f4a7c15ULL + (h << 6) + (h
>> 2);
        return h;
    }
};
std::unordered_map<Product, double, ProductHash> price;
price[{"north", 7}] = 101.5;
```

CONTAINER	KEY MUST PROVIDE	RULE
`map`, `set`	a **strict weak ordering** (`operator<` or a comparator)	irreflexive and transitive; equivalence means "neither is less". `≤` is **not** valid: undefined behaviour
`unordered_map`, `unordered_set`	a hash and `operator==`	equal keys **must** have equal hashes; a poor hash makes every lookup O(n)

- **Transparent comparators:** `std::map<std::string, int, std::less<>>` lets `find` take a `string_view` or `const char*` without building a temporary `std::string` (compiled: it found `"apple"` from a `string_view`). C++20 adds the same for unordered containers.
- **Never change a key** while it is in the container: its position or bucket would be wrong. Keys are `const` for that reason.
- **Hash quality:** combine *all* fields; avoid returning just one field. `std::hash` for `int` is the identity in libstdc++, which is fine.
- A `rehash` (growth beyond `max_load_factor`) invalidates iterators; `reserve(n)` up front avoids it, and avoids the allocations on a hot path.

- Ordered or hashed? Ordered when you need sorted iteration or range queries (`lower_bound`); hashed for plain lookup. For a few dozen keys a sorted `vector` and binary search often beats both.

> KEY POINT "How do you use your own type as a key in `unordered_map`?" Provide `operator==` and a hash functor (or specialise `std::hash`), combining all members, and make sure equal objects hash equally.

`pair`, `tuple` and structured bindings



map::insert returns pair<iterator, bool>: structured bindings read it as [it, inserted].

A pair returns two values without a throwaway struct.

`std::pair<A, B>` holds two values, `std::tuple<A, B, C ...>` any number. Structured bindings (C++17) unpack them into named variables.

```
// returns {smallest, largest}
std::pair<int, int> minmax(const std::vector<int>& v);
// lo = 1, hi = 9: two names instead of .first / .second
auto [lo, hi] = minmax({3, 9, 1, 7});

auto t = std::make_tuple(1, 2.5, std::string("x"));
// 2.5 (index must be a compile-time constant)
std::get<1>(t);
// "x" (or by type, when the type appears once)
std::get<std::string>(t);

std::map<std::string, int> m{{"a", 1}};
// insert returns pair<iterator, bool>
auto [it, inserted] = m.insert({"a", 2});
// inserted == false, it->second == 1: the key existed, nothing changed
```

- A pair or tuple has **no names**: `.first` and `std::get<2>` say nothing about meaning. For a result that leaves a function and is used in more than one place, a small named `struct` reads better.
- Comparison is built in and lexicographic: pairs and tuples compare element by element, so they work as `map` keys and in `std::sort`.
- `std::tie(a, b) = f();` assigns into existing variables; `std::ignore` skips one.
- Structured bindings also unpack plain structs and arrays: `auto [x, y] = point;`. They are copies unless you write `auto&` or `const auto&`.

Further reading by section

Where each section of this lesson is explained in more depth. Pages are PDF viewer pages (the number your PDF reader shows).

SECTION	BOOK	PART OF THE BOOK	PDF PAGES
Associative and unordered containers	C++ Primer	11.1 to 11.2 Using an associative container, overview	449 to 456
	C++ Primer	11.3 Operations on associative containers	457 to 471
	Programming: Principles and Practice	20.1 to 20.6 map, unordered_map, set, container overview	591 to 609
Custom keys: comparators and hashes	C++ Primer	11.3 Operations on associative containers	457 to 471
	C++ Primer	11.4 The unordered containers	472 to 477
pair, tuple and structured bindings	C++ Primer	17.1 The tuple type	747 to 751